

PYTHON ENGINEERING SENSOR DATA ANALYSIS TOOL

1. OVERVIEW

Engineering systems often generate large volumes of measurement data from sensors and monitoring equipment. Analyzing this data efficiently is essential for understanding system performance, identifying trends, and detecting abnormal operating conditions.

This project demonstrates how Python can be used as a lightweight engineering tool to analyze time-series sensor data exported from industrial systems. The workflow focuses on transforming raw measurement data into structured insights through automated data processing, statistical analysis, and visualization.

The analysis pipeline reads measurement data from Excel, performs basic statistical analysis, and generates visualizations of key operational parameters such as temperature, pressure, and flow rate. In addition, a simple anomaly detection method based on deviation from a moving average is implemented to highlight potential irregularities in system behavior.

The objective of this example is to illustrate how engineers can quickly build small analytical tools to support system monitoring, diagnostics, and operational analysis without requiring complex software platforms. Such tools can be particularly useful during system commissioning, troubleshooting, or performance evaluation.

This example uses a simplified dataset representing sensor measurements collected over time, but the same approach can be applied to real-world industrial data from process systems, energy systems, or automation environments.

2. INPUT DATA

The dataset represents time-series measurements collected from an industrial system. The following parameters are included:

- Time (s) – measurement timestamp
- Temperature (°C) – system temperature reading
- Pressure (bar) – system pressure
- Flow (L/min) – system flow rate

3. DATA PROCESSING

The analysis script performs the following steps:

- Import sensor data from Excel
- Perform statistical analysis of measurements
- Generate visualizations of system behavior

- Detect simple anomalies using deviation from moving average
- Export results and summary statistics

4. RESULTS AND VISUALIZATIONS

The measurement data was analyzed using Python to identify trends and operational characteristics of the system. Time-series visualizations were generated to provide a clear view of how key parameters evolve during operation.

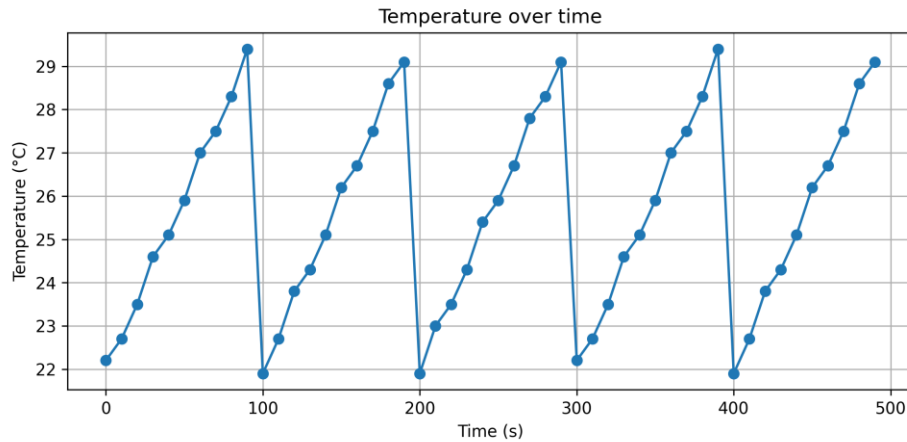


Figure 1 - Temperature trend over time

The temperature plot shows the variation of system temperature over time. This visualization helps identify general thermal trends and potential deviations from expected operating conditions. The moving average curve provides a smoothed representation of the temperature profile, making it easier to observe longer-term trends while reducing the effect of short-term measurement fluctuations.

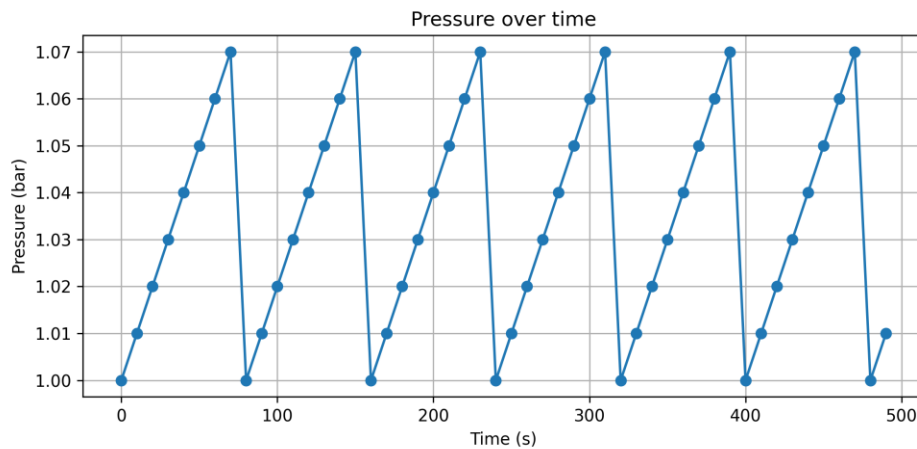


Figure 2 - Pressure behavior during operation

Pressure data was visualized as a function of time. This allows engineers to observe system pressure stability and identify potential transient behavior. Consistent pressure values typically indicate stable system operation, while sudden changes could signal disturbances or system adjustments.

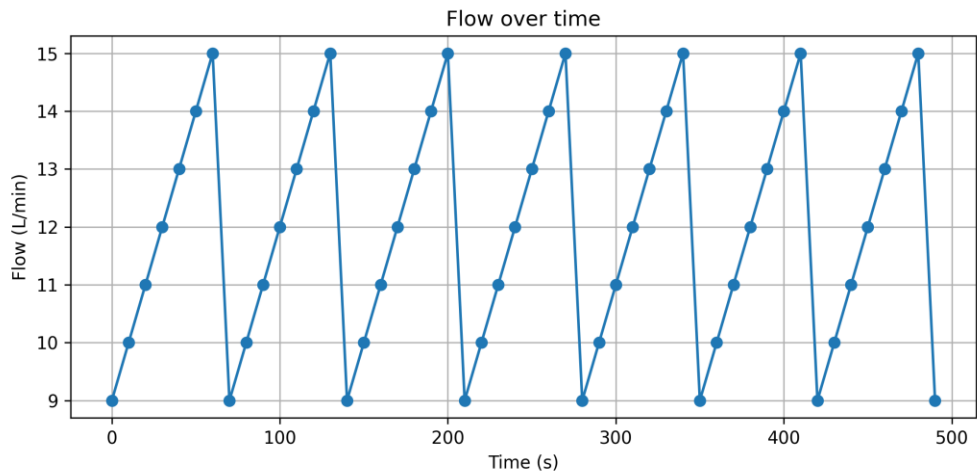


Figure 3- Flow rate variation

Flow rate measurements were analyzed to understand how system throughput changes during operation. The visualization highlights variations in flow that may be related to changes in system demand or control adjustments.

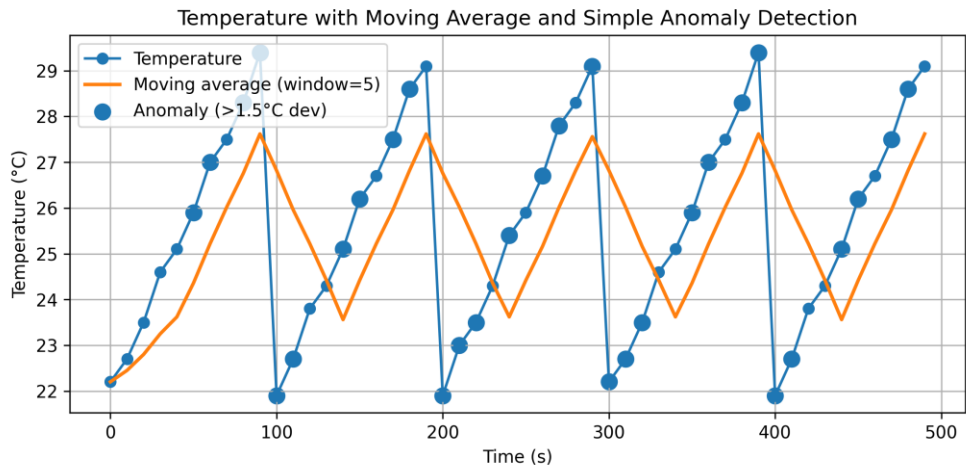


Figure 4 - Temperature moving average and anomaly detection

In addition to the basic visualizations, a simple anomaly detection approach was applied to the temperature data. By comparing the measured temperature values to a moving average, the analysis highlights data points that deviate significantly from expected behavior. This technique can help engineers quickly identify unusual operating conditions or sensor anomalies.

Together, these visualizations demonstrate how simple data processing and visualization techniques can provide valuable insights into system performance and operational stability.

5. TOOLS AND TECHNOLOGIES

- Python
- Pandas (data processing)
- Matplotlib (data visualization)
- Microsoft Excel (data preparation)
- Google Colab (execution environment)

6. ENGINEERING INSIGHTS

The analysis demonstrates how simple data processing techniques can provide insight into system performance trends. Visualization helps engineers identify operational patterns, while anomaly detection can highlight potential issues in system behavior.

This example demonstrates how small Python tools can support engineering analysis workflows by transforming raw measurement data into actionable insights.

7. POTENTIAL APPLICATIONS

The same approach can be applied to a wide range of engineering and industrial scenarios, including:

- Monitoring sensor data from industrial processes
- Analyzing measurement data from test systems
- Detecting anomalies in operational parameters
- Generating automated reports for engineering teams
- Supporting system commissioning and troubleshooting

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